

Double Pulse Test Circuit

Why need Double Pulse Testing?

- Gate drive functionalities.
- Switching energy losses.
- Voltage overshoots across DUT(device under test).
- Device characteristics at different case temperatures.

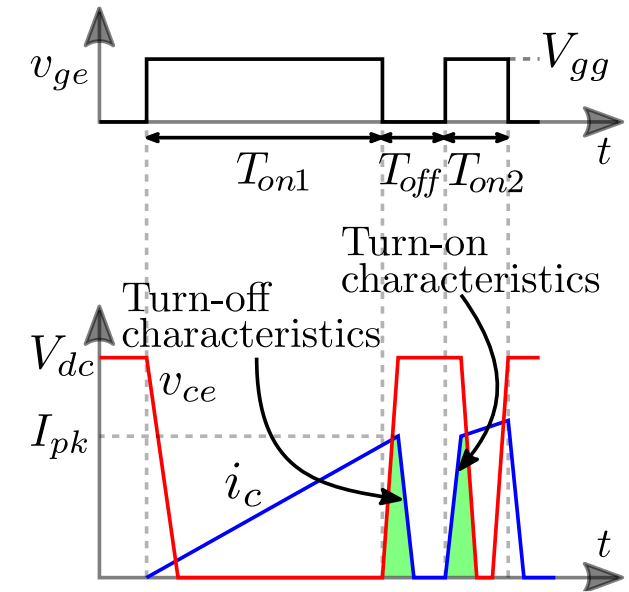
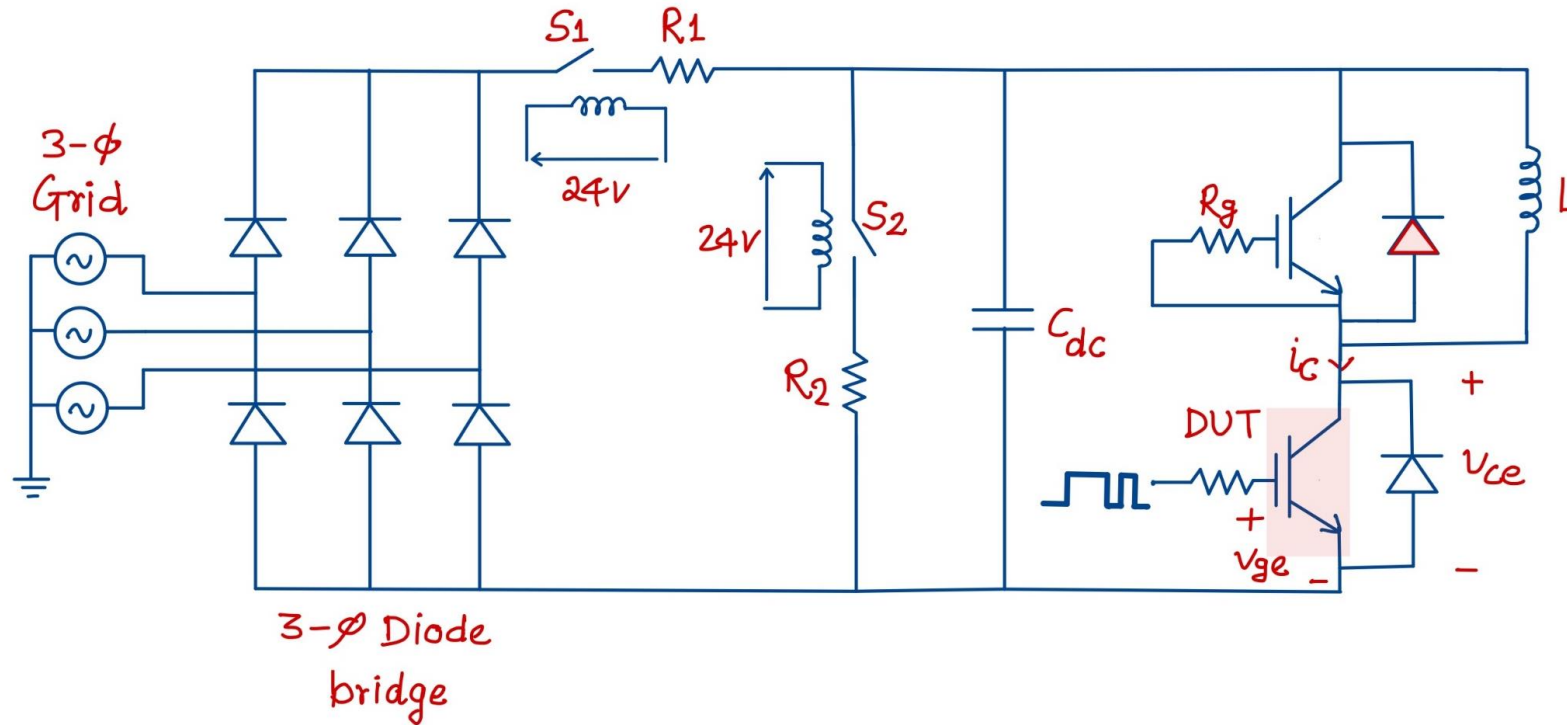
An Example: IKW15N120T2 Datasheet

Switching Characteristic, Inductive Load, at $T_j=25^\circ\text{C}$

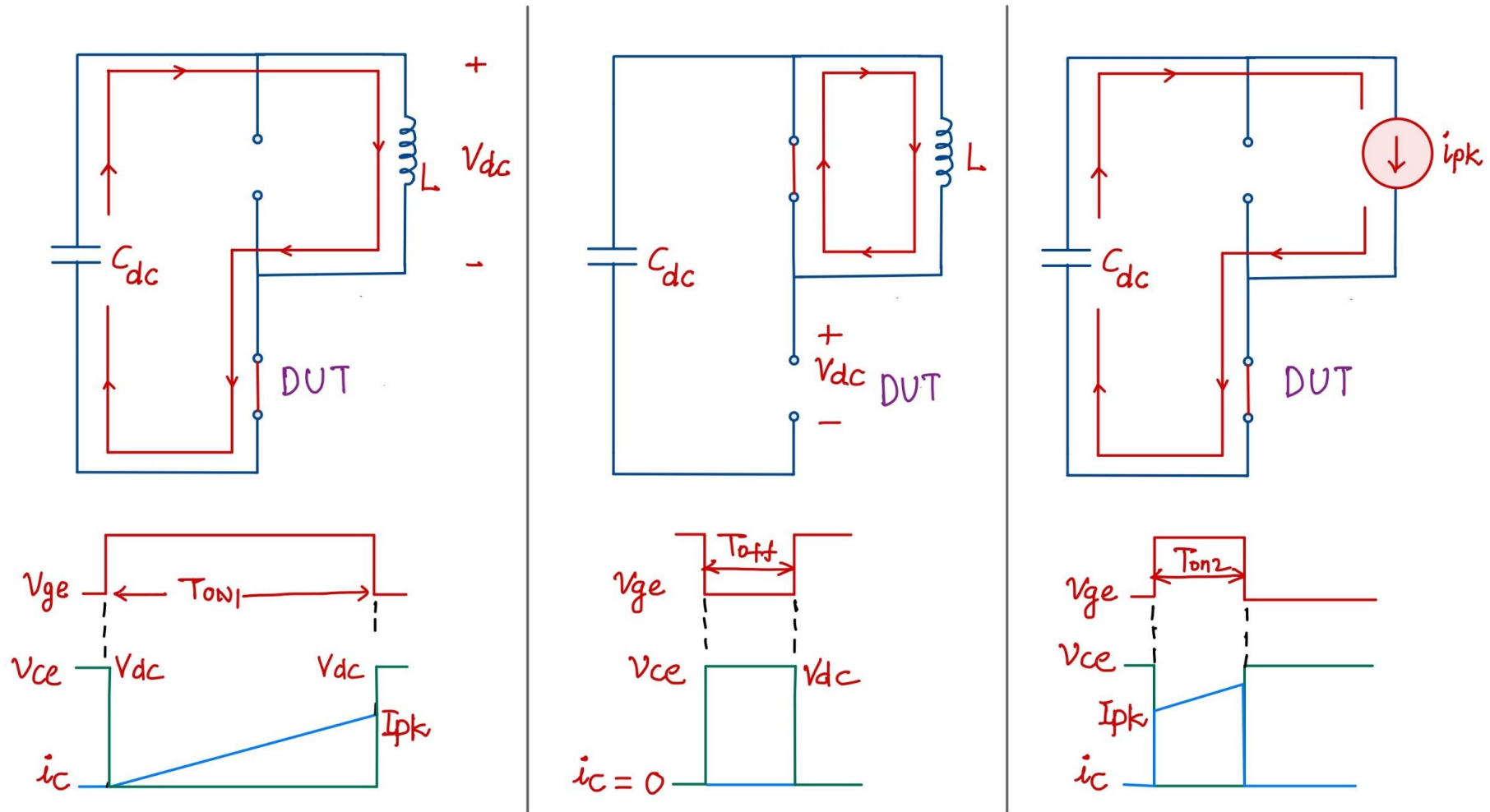
Parameter	Symbol	Conditions	Value			Unit
			min.	typ.	max.	
IGBT Characteristic						
Turn-on delay time	$t_{d(on)}$	$T_j=25^{\circ}\text{C}$, $V_{CC}=600\text{V}$, $I_C=15\text{A}$, $V_{GE}=0/15\text{V}$, $R_G=41.8\Omega$, $L_{\sigma}^{2)}=126\text{nH}$, $C_{\sigma}^{2)}=34\text{pF}$ Energy losses include “tail” and diode reverse recovery.	-	32	-	ns
Rise time	t_r		-	25	-	
Turn-off delay time	$t_{d(off)}$		-	362	-	
Fall time	t_f		-	95	-	
Turn-on energy	E_{on}		-	1.25	-	mJ
Turn-off energy	E_{off}		-	0.8	-	
Total switching energy	E_{ts}	-	2.05	-		
Anti-Parallel Diode Characteristic						
Diode reverse recovery time	t_{rr}	$T_j=25^{\circ}\text{C}$, $V_R=600\text{V}$, $I_F=15\text{A}$, $di_F/dt=450\text{A}/\mu\text{s}$	-	300	-	ns
Diode reverse recovery charge	Q_{rr}		-	1.3	-	μC
Diode peak reverse recovery current	I_{rrm}		-	10	-	A
Diode peak rate of fall of reverse recovery current during t_b	di_{rr}/dt		-	215	-	$\text{A}/\mu\text{s}$

DPT setup enables to characterize the device at user specific conditions.

DPT Circuit Diagram and Test Procedure



Equivalent circuits during T_{on1} , T_{off} and T_{on2} :



I_{pk} and V_{DC} are user-defined values.

Calculation of L and C parameters:

$$L > \frac{V_{DC} T_{on2}}{0.05 I_{pk}}$$

$$C_{DC} > \frac{\Delta Q}{\Delta V_{DC}}$$

$$\Delta Q = \frac{1}{2} T_{on1} I_{pk} + I_{pk} T_{on2};$$

$$\Delta V_{DC} = 1\% V_{DC}$$

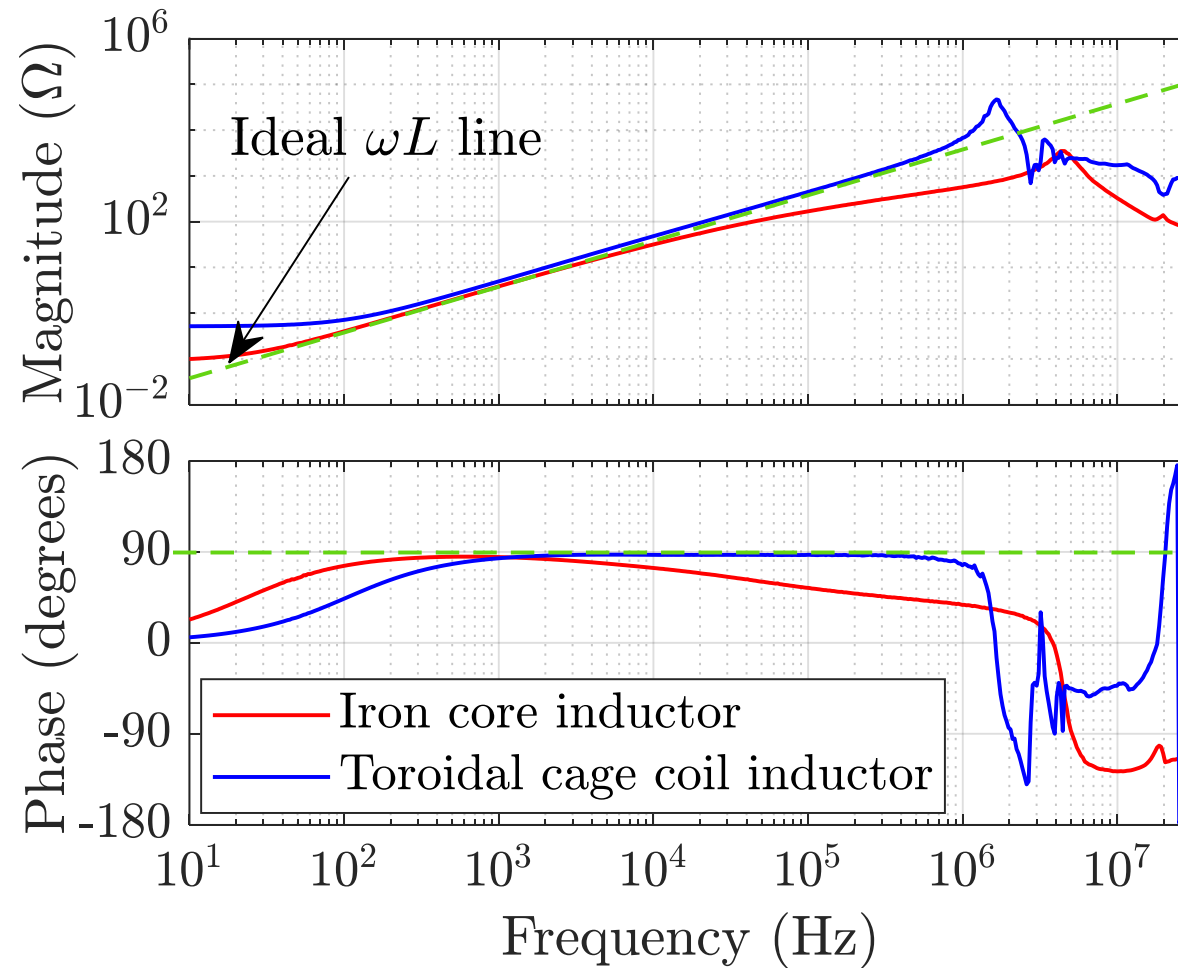
Calculation of T_{on1} , T_{off} and T_{on2} :

$$T_{on1} = \frac{LI_{pk}}{V_{DC}}$$

$$T_{on2} = T_{off} = 2 \sim 3 \mu s$$

- T_{on2} : To limit the current.
- T_{off} : Inductor stored energy should not completely dissipate into FWD resistance.

Choice of Inductor:

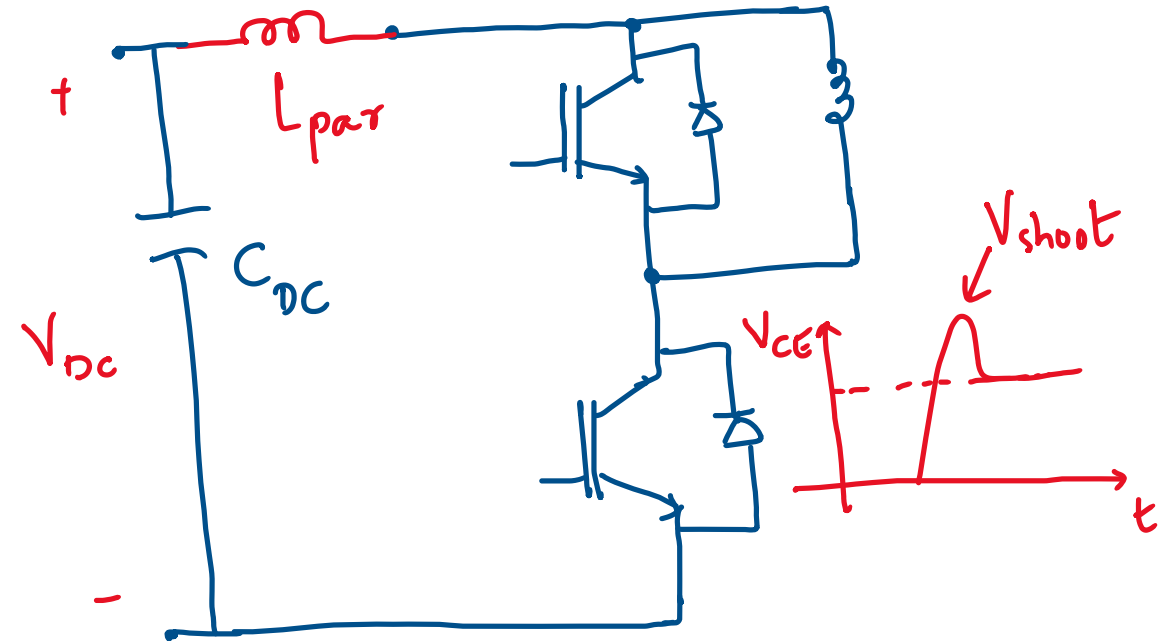


Air core inductor is preferred due to its better linearity w.r.t frequency.

DC Bus Capacitor Formation:

Table 1: Capacitor Bank Approach Comparison		
Parameter	Large Capacitor Approach	Parallel Array Approach
Capacitance	70 μ F	72 μ F
Voltage Rating	1.2kV	1.4kV
ESL	22nH	6.67nH
Cost (1k)	\$204.16	\$114.48

$$* L_{par} = L_{ESL} + L_{PCB}$$



Parallel array approach is preferred.

Reference:

- B. Mondal, R. T. Pogulaguntla and A. K. B, "Double Pulse Test Set-up: Hardware Design and Measurement Guidelines," *2022 IEEE International Conference on Power Electronics, Drives and Energy Systems (PEDES)*, Jaipur, India, 2022, pp. 1-6, doi: 10.1109/PEDES56012.2022.10080339.

